Absolutely! Below is your **complete, professionally formatted Requirement Gathering Document**, **including the plain text version of the eauto database schema**, ready for official documentation or submission.

🛠️ Requirement Gathering Document

Project Title: eAuto – Bike Spare Parts E-Commerce Platform

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1. Project Overview

The goal of the **eAuto** project is to build a responsive, user-friendly, and scalable e-commerce platform that enables users to browse, search, and purchase spare parts for bikes from multiple brands. The system will support user authentication, shopping cart functionality, order management, and a full-featured admin panel for product and order control.

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2. Functional Requirements

2.1 User Authentication

- User registration and secure login/logout
- "Remember Me" functionality
- Forgot password/reset mechanism
- Session-based access management

2.2 Product Catalog

- Display products grouped by brand (e.g., Hero, Honda, Brand B)
- Product card with:
    - Product image
    - Name
    - Brand
    - Price and discount
    - Availability (In Stock / Sold Out)
    - “Add to Cart” button

2.3 Shopping Cart

- Add products to cart (via PHP form)

- View cart in dropdown and on dedicated page
- Update quantity or remove items
- Dynamic price calculation
- Disable checkout when cart is empty

2.4 Checkout Process

- Checkout form with address details:
    - First Name, Last Name
    - Phone Number
    - Company (optional)
    - Address, City, State/Province, ZIP Code, Country
    - “Set as Default” address option
- Place order and receive confirmation

2.5 User Dashboard

- View and edit user profile
- Manage saved addresses
- View order history with status

2.6 Admin Panel

- Secure admin login
- Add, edit, and delete products
- Upload and manage product images
- View and manage customer orders
- View user details and addresses

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3. Non-Functional Requirements

3.1 Usability

- Fully responsive design (mobile, tablet, desktop)
- Intuitive navigation using navbar and sidebar

- Smooth animations using AOS (Animate on Scroll)

3.2 Performance

- Optimized MySQL queries for speed
- Lazy loading for product images
- Efficient session and cart handling

3.3 Security

- Session-based user/admin authentication
- SQL injection protection using prepared statements
- Access control for admin features

3.4 Testing

- Manual testing for all user flows
- Cross-browser compatibility (Chrome, Firefox, Edge, Safari)

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4. Database Design (MySQL – eauto)

Key Tables

Table Name	Description
users	User credentials and profile data
products	Product listings with brand/category details
cart	Temporary cart data for each user
orders	Order master records
order_items	Individual items within an order
admins	Admin login credentials

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5. User Interface Components

5.1 Header

- Top banner with logo
- Search bar for products
- Cart icon with item count
- User dropdown (Profile, Logout)

5.2 Sidebar

- Collapsible category and brand filter menu
- Visible on larger screens, toggle on mobile

5.3 Product Card

- Image, title, brand, price, discount
- Stock status label
- “Add to Cart” button with form submission

5.4 Cart Dropdown

- Shows product name, quantity controls, and total price
- Empty state message when no items are present

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6. Technology Stack

Layer	Technology
Frontend	HTML, CSS, Bootstrap 5, JavaScript, AOS
Backend	PHP
Database	MySQL (phpMyAdmin)
Server	XAMPP (Apache + MySQL)
Admin Panel	PHP-based dashboard

Layer	Technology
Image Uploads	File input + PHP handler

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7. Optional Future Enhancements

- Live product search with auto-suggestions
- Payment gateway integration (Razorpay, Stripe, etc.)
- Order tracking system
- Product reviews and star ratings
- Inventory and stock management
- Email or SMS order confirmations

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8. Conclusion

This document outlines the essential requirements for developing the **eAuto** bike spare parts platform. It serves as a base for design, development, and testing, ensuring alignment between technical teams and stakeholders.

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9. Plain Text Database Schema

Database Name: eauto

Table: admins

- id – INT, AUTO_INCREMENT, PRIMARY KEY
- email – VARCHAR(100), NOT NULL, UNIQUE
- password – VARCHAR(255), NOT NULL
- created_at – TIMESTAMP, default CURRENT_TIMESTAMP

Table: users

- id – INT, AUTO_INCREMENT, PRIMARY KEY

- fullname – VARCHAR(255), NOT NULL
- email – VARCHAR(255), NOT NULL, UNIQUE
- contact – VARCHAR(20)
- birthdate – DATE
- gender – VARCHAR(10)
- password – VARCHAR(255), NOT NULL

Table: products

- id – INT, AUTO_INCREMENT, PRIMARY KEY
- product_name – VARCHAR(255), NOT NULL
- price – DECIMAL(10,2), NOT NULL
- original_price – DECIMAL(10,2)
- discount_amount – DECIMAL(10,2)
- image – VARCHAR(255)
- brand – VARCHAR(100)
- main_brand – ENUM('BRAND B', 'Hero', 'Honda'), NOT NULL
- rating – FLOAT
- reviews – INT
- stock_status – ENUM('In Stock', 'Sold Out'), DEFAULT 'In Stock'

Table: cart

- id – INT, AUTO_INCREMENT, PRIMARY KEY
- product_name – VARCHAR(255), NOT NULL
- price – DECIMAL(10,2), NOT NULL
- image – VARCHAR(255), NOT NULL
- brand – VARCHAR(255), NOT NULL
- quantity – INT, DEFAULT 1

Table: orders

- id – INT, AUTO_INCREMENT, PRIMARY KEY
- user_id – INT (foreign key to users.id)
- fullname – VARCHAR(255)
- email – VARCHAR(255)
- phone – VARCHAR(20)
- address – TEXT
- city – VARCHAR(100)
- state – VARCHAR(100)
- zip – VARCHAR(20)
- total_amount – DECIMAL(10,2)
- payment_method – VARCHAR(50)
- created_at – TIMESTAMP, DEFAULT CURRENT_TIMESTAMP

Table: order_items

- id – INT, AUTO_INCREMENT, PRIMARY KEY
- order_id – INT (foreign key to orders.id)
- product_name – VARCHAR(255)
- brand – VARCHAR(100)
- quantity – INT
- price – DECIMAL(10,2)
- product_image – VARCHAR(255)

 Admin Panel Files:

File Name	Purpose
admin-customers.php	Manage or view registered customers
admin-dashboard.php	Admin home with site stats
admin-orders.php	Manage all orders
admin-sidebar.php	Sidebar menu layout
delete-product.php	Delete products from DB
edit-product.php	Edit existing product details
export-products.php	Export product list (CSV/Excel)
login.php	Admin login
logout.php	Logout admin
manage-products.php	Add/view/update/delete products
print-label.php	Generate/print shipping/product labels
register_admin.php	Register a new admin

🐘 PHP Files (Main Application)

File Name	Purpose
add_to_cart.php	Backend logic to add selected products to the user's cart
Bajaj.php	Product listing page for Bajaj bike spare parts
cart.php	Displays the shopping cart (items added, quantity, total, etc.)
checkout.php	Checkout page to place an order and enter delivery details
checkout-action.php	Processes the order after checkout form submission
config.php	Database configuration file (DB host, user, password, name)
delivery-address.php	Form to enter or update user's shipping/delivery address
footer.php	Common footer included across all pages
header.php	Common header with navbar, cart icon, search bar, etc.
Hero.php	Product listing page for Hero bike spare parts
Honda.php	Product listing page for Honda bike spare parts
index.php	Homepage of the website (introduction, banners, top products)
Login.php	Login form for users to access their profile/cart
login-action.php	Validates login credentials and starts user session
logout.php	Destroys session and logs the user out

File Name	Purpose
my-profile.php	User profile page showing personal info and address
order-details.php	Displays detailed view of a specific order (items, price, status)
orders.php	Lists all orders placed by the user
order-success.php	Success page shown after placing an order
registration.php	User registration page to create an account
